Orthopaedic Conditions of the Hip — Adults (incl. Neglected DDH)

The Big Picture

This adult-hip lecture revisits **AVN**, **hip OA**, and **THR** (same core as the earlier hip lecture) and adds the key new topic: **neglected developmental dysplasia of the hip (DDH) in adults**. Focus your revision on the **DDH** section — the rest is recap.

Quick Recap — AVN, OA, THR

- **AVN** — ↓ blood flow → femoral head death → collapse. Male 35–50, **bilateral 80%**. Causes: **steroids/alcohol** (fat-cell hypertrophy blocks vessels), trauma (medial femoral circumflex), sickle cell, idiopathic (~20%). **Steinberg/Ficat** staging; **crescent sign (III) = collapse**. **MRI 99%**. Treatment ladder: bisphosphonate (pre-collapse) → **core decompression** → free fibula → **THR** (collapse/>40/irreversible).
- **Hip OA** — cartilage loss; **groin pain + limited internal rotation**; **Tonnis** grading; X-ray narrowing/osteophytes/sclerosis/cysts. Non-op first (NSAID, **stick in opposite hand**, weight loss, steroid), then **THA (>50, end-stage)**.
- **THR** — **Charnley low-friction** (metal head + poly cup + cement); **~80% at 25 yrs** ("operation of the century"). **Sepsis = chief contraindication.** Fixation press-fit vs cemented; bearings metal/ceramic/poly. Complications: acute (dislocation, **sciatic palsy**, periprosthetic fracture), chronic (**aseptic loosening**).

Neglected DDH in Adults (the new material)

- **Definition** — DDH untreated into adulthood → anatomical distortion, pain, gait abnormality; usually presents as **secondary OA**.
- **Epidemiology** — **female > male**, high in **swaddling** cultures; many neglected (mild early symptoms).
- **Pathoanatomy** — **shallow acetabulum**, **high-riding femoral head**, soft-tissue contractures, **limb-length discrepancy**, narrow femoral canal.
- **Clinical** — **Trendelenburg gait + positive Trendelenburg sign**, limp, pain, limited ROM, leg-length discrepancy. Most seek help for **hip pain + limp**.
- **Classification** — **Crowe (I–IV)**, **Hartofilakidis (dysplasia / low dislocation / high dislocation)** — guide surgical planning.
- **Imaging** — AP pelvis X-ray (**shallow acetabulum + superior femoral head migration**); CT for bony anatomy; MRI for cartilage/labrum in young adults.
- **Management** — non-operative (analgesia, physio) only temporises; definitive = **THR**, often with **femoral shortening osteotomy** in severe high dislocation.
- **THR challenges** — high hip centre, deficient/small acetabulum, narrow femoral canal → need **modular stems** / subtrochanteric osteotomy.
- **Complications** — **sciatic nerve palsy** (esp. with lengthening — reason for femoral shortening), dislocation, leg-length discrepancy, intraoperative fracture.

SBA answers from lecture

- Commonest sign in neglected DDH = **Trendelenburg gait**.
- Characteristic X-ray = **shallow acetabulum + superior migration**.

- Nerve most at risk in THR for severe DDH = **sciatic nerve**.
- Purpose of femoral shortening osteotomy = **prevent sciatic nerve stretch**.
- Usual reason adults present = **hip pain + limp**.

Walk-Away Points

- **Neglected DDH** = female-predominant, presents as secondary OA with **Trendelenburg gait + leg-length discrepancy**.
- **Crowe / Hartofilakidis** classify it; X-ray = **shallow acetabulum + high-riding head**.
- Definitive treatment = **THR ± femoral shortening osteotomy** (protects the **sciatic nerve**).
- AVN/OA/THR recap: crescent sign → collapse → THR; **sepsis = THR's chief contraindication**.

Hip — Adults (incl. Neglected DDH) — Revision Table

Neglected DDH	Detail
Definition	Untreated DDH → adult; presents as **secondary OA**
Epidemiology	**Female > male**, swaddling cultures
Pathoanatomy	Shallow acetabulum, **high-riding head**, narrow canal, LLD
Clinical	**Trendelenburg gait + sign**, limp, pain, LLD
Presenting complaint	**Hip pain + limp**

DDH classification	Detail
Crowe	I–IV
Hartofilakidis	Dysplasia / low / high dislocation

DDH imaging	Finding
X-ray	**Shallow acetabulum + superior migration**
CT	Bony anatomy
MRI	Cartilage/labrum (young)

DDH management	Detail
Non-op	Analgesia, physio (temporary)
Definitive	**THR ± femoral shortening osteotomy**
THR challenges	High hip centre, deficient acetabulum, modular stems
Nerve at risk	**Sciatic** (osteotomy prevents stretch)
Complications	Sciatic palsy, dislocation, LLD, fracture

Recap	Key point
AVN	Crescent sign (III) = collapse; steroid/alcohol; MRI 99%
Hip OA	Groin pain, ↓ internal rotation; Tonnis; THA >50
THR	**Charnley**, 80%/25yr; **sepsis = chief CI**; aseptic loosening

Bold = high-yield.